Digital Electronics Lab Report

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Lab code : EE29210

Date of experiment : 13-1-26

Experiment No : 1

7-segment LED display and IC-7447 (BCD-to-7-segment decoder/driver)

Objectives

- To identify pins of 7-segment (common-anode) light-emitting diode (LED) display
- To understand functions of pins of IC 7447 and use it to operate the LED display
- Blanking the leading zeroes in a multi-digit display.
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Apparatus:

- IC-7447 -1

- 7-segment LED displays - Common anode
- Resistor - 470 KΩ or 390 KΩ - 8
- Resistor 2 KΩ - 4
- Breadboard
- DIP Switch With 4 or more switches - 1
- Connecting wires
- DC Power Supply 5V – 1

Theory:

A 7-segment LED display is a commonly used electronic display device for representing decimal digits (0–9).

It consists of seven LED segments, named a, b, c, d, e, f, g, arranged in a specific pattern.

By selectively glowing different segments, numerical digits can be displayed.

There are two types of 7-segment LED displays:

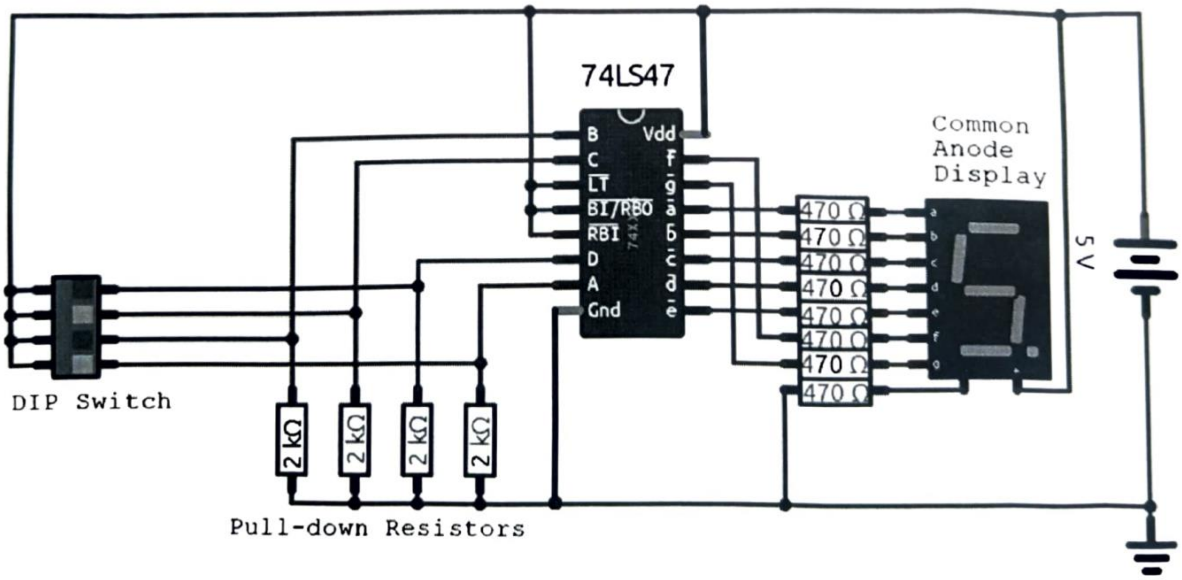
- **Common Anode**
- **Common Cathode**

In a common-anode 7-segment display, all the anodes of the LED segments are connected together and tied to the positive supply (+5 V). A segment glows when its cathode is driven LOW, allowing current to flow through the LED.

The BCD input is provided using a DIP switch. Based on the BCD value, IC 7447 activates the required segment outputs by pulling them LOW.

Current flows from the common anode through the LED segment and the current limiting resistor into the IC output causing the segment to glow and display the corresponding decimal digit.

<u>Circuit diagram:</u>



<u>Pin Functions of IC 7447</u>

I. **BCD Inputs (A, B, C, D):**
These inputs accept a 4-bit BCD number corresponding to decimal digits 0–9.

II. **Segment Outputs (a–g):**
These outputs drive the corresponding segments of the 7-segment display. A LOW output turns ON the segment.

III. **Lamp Test (LT):**
When LT = 0, all segments glow irrespective of the BCD input. It is used to check whether all LED segments are functioning properly.

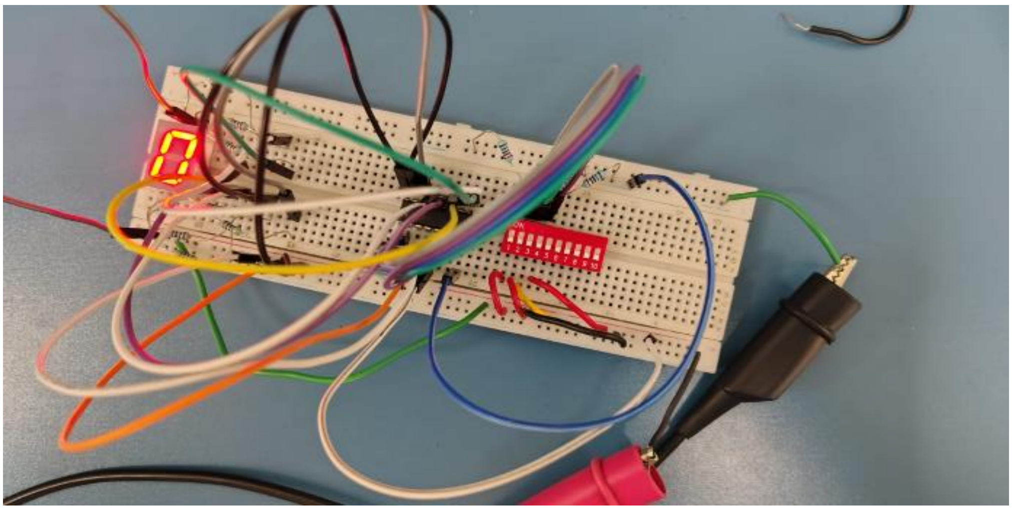
IV. **Ripple Blanking Input (RBI):**
Used for leading zero suppression. When RBI is LOW and the BCD input is zero, the display is blanked.

V. **Ripple Blanking Output (RBO):**
Used in multi-digit displays. RBO of a less significant digit is connected to RBI of the next significant digit to suppress leading zeros.

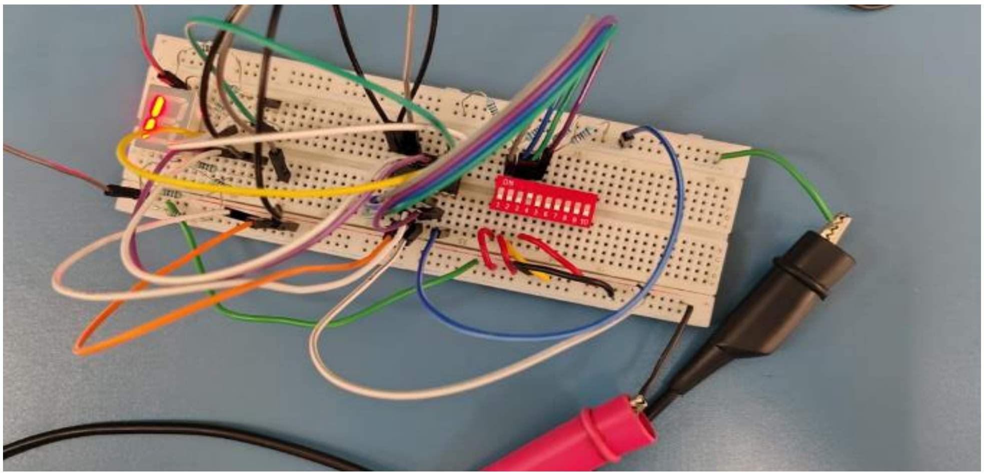
VI. **Blanking Input (BI/RBO):**
When BI is LOW, the entire display is turned OFF.

Observation:

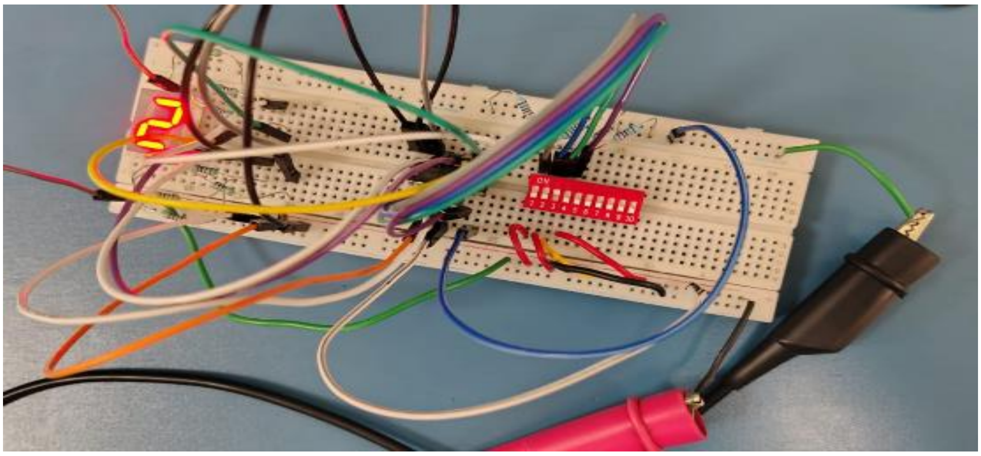
Combination for 0:



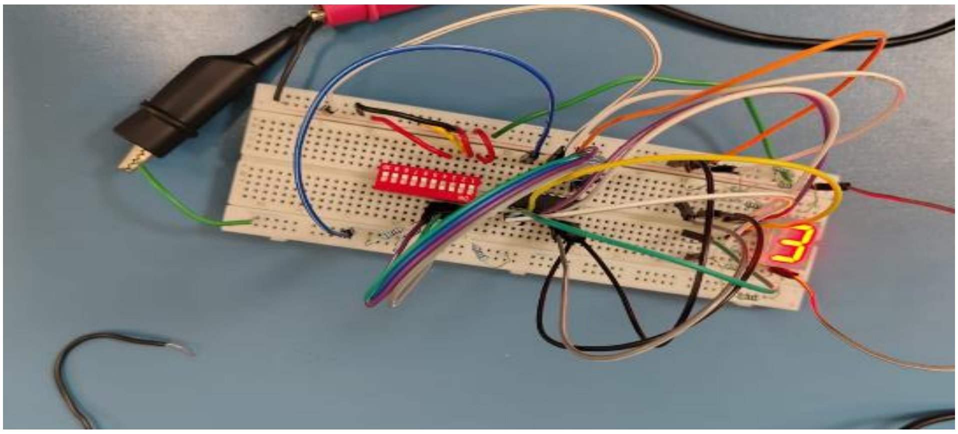
Combination for 1:



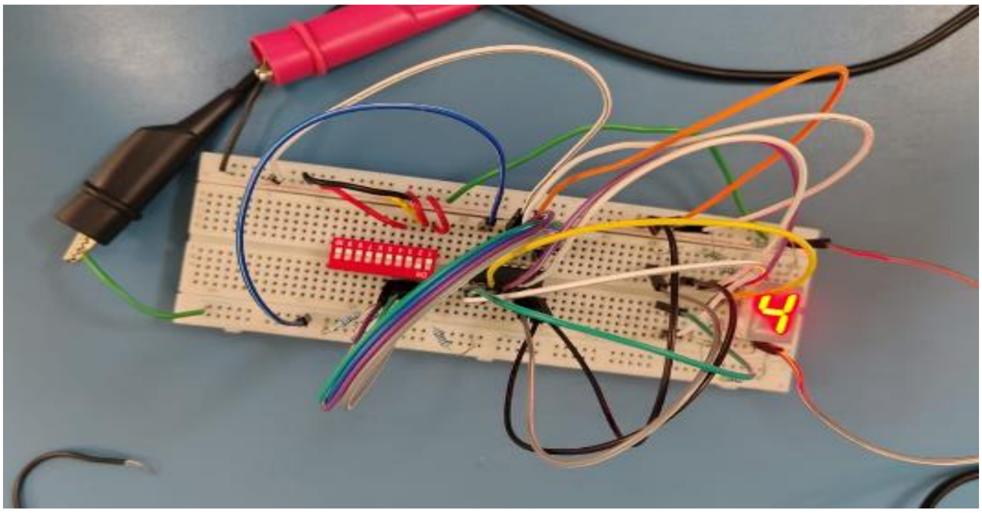
Combination for 2:



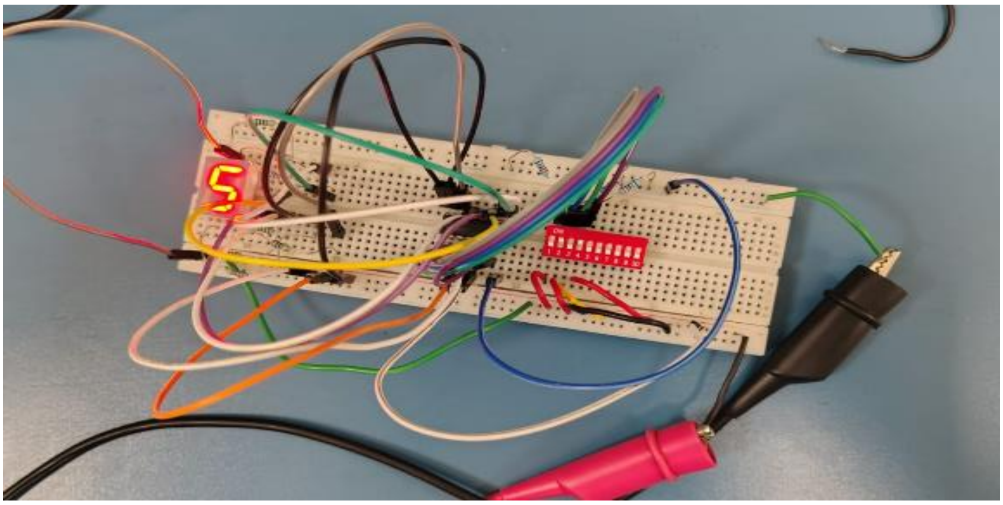
Combination for 3:



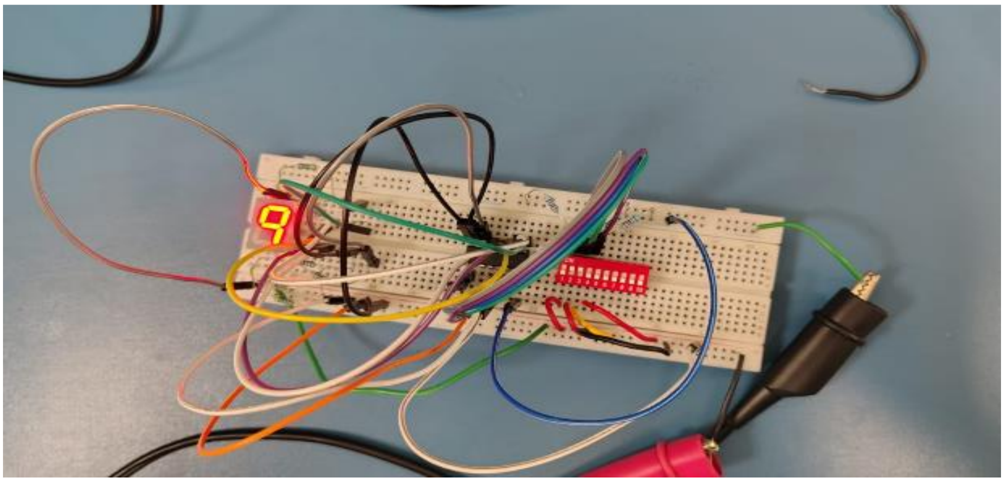
Combination for 4:



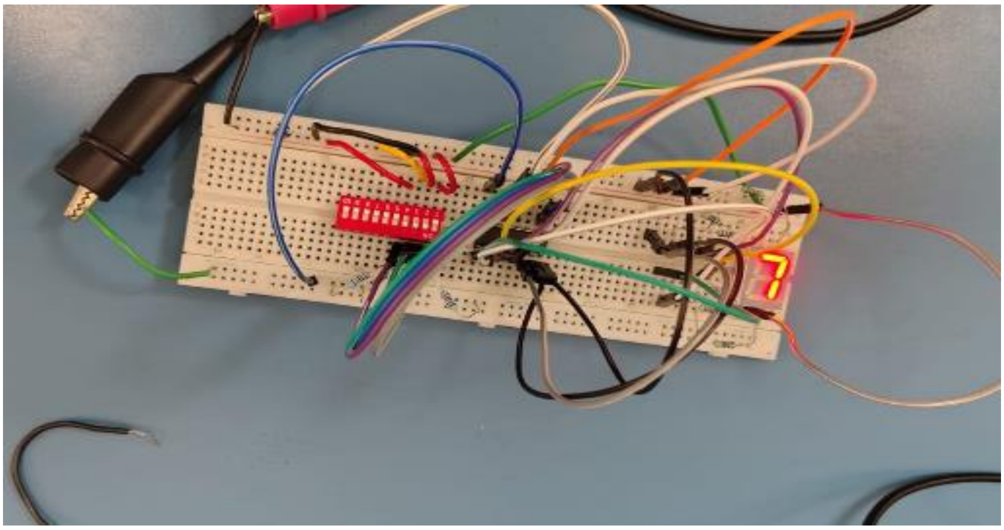
Combination for 5:



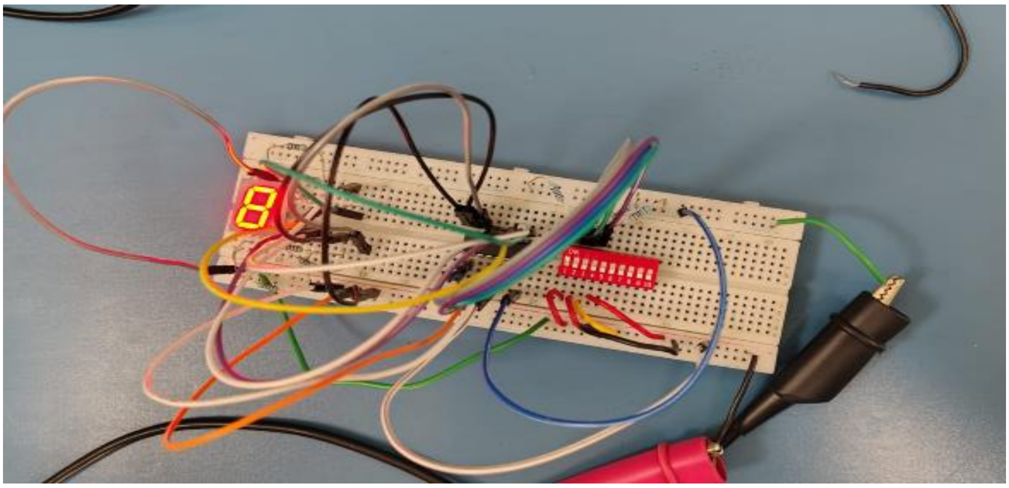
Combination for 6:



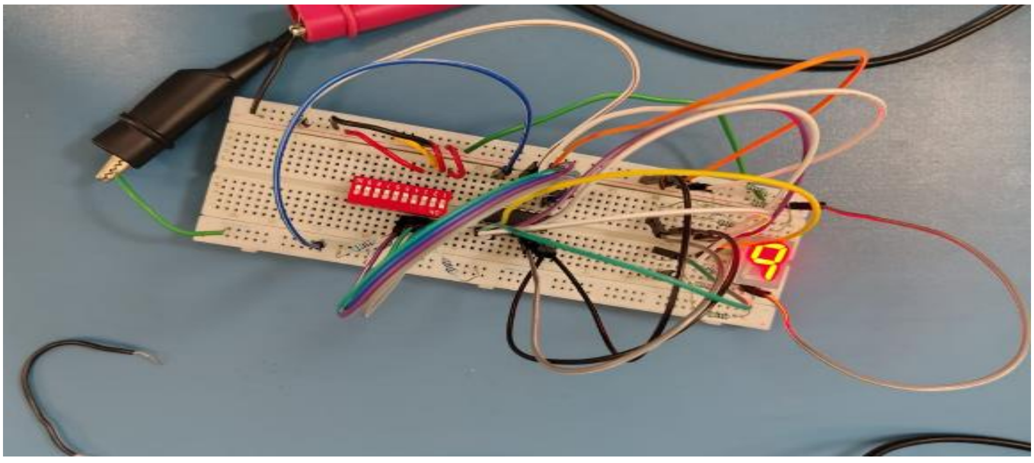
Combination for 7:



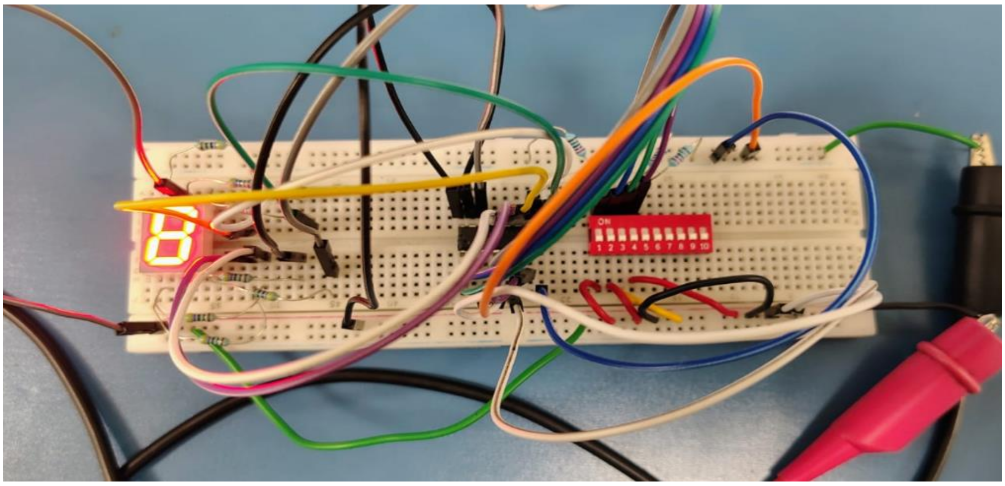
Combination for 8:



Combination for 9:

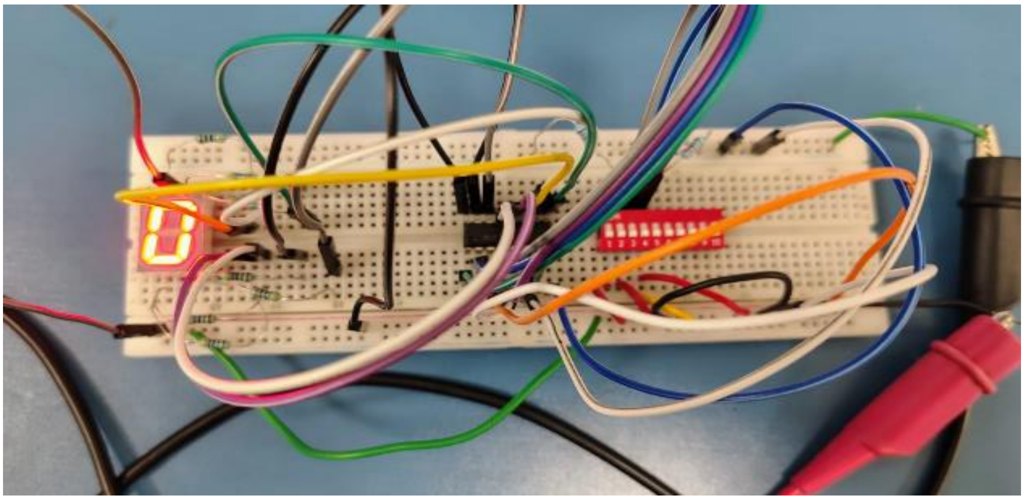


LT = 0V, RBI = 5V, RBO = 5V :

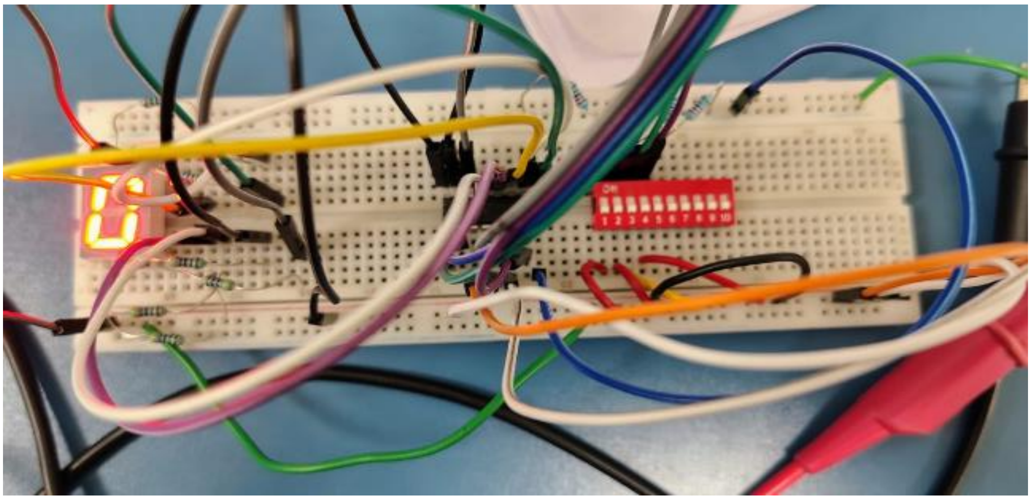


LT = 5V, RBO = 5V and DCBA = 0000

a. RBI = 0V:



b. RBI=5V:



- It is observed that the IC 7447 correctly decodes the BCD input (0000–1001) and activates the appropriate segments of the common-anode 7-segment display.
- The display shows the correct decimal digits 0 to 9, confirming the proper functioning of the BCD-to-7-segment decoding operation.
- The lamp test (LT) input turns ON all segments
- The ripple blanking feature successfully suppresses leading zeros

<u>Result:</u>

It is observed that the IC 7447 correctly decodes BCD inputs (0000–1001) and displays the corresponding decimal digits on the common-anode 7-segment display, confirming proper BCD-to-7-segment decoding.

The active-LOW nature of the output pins is verified, as the segments glow when the corresponding outputs are LOW.

The ripple blanking feature functions correctly, as leading zeros are suppressed when appropriate input conditions are applied.

However, during Step 3(a), the expected operation was not observed, which may be due to improper pin connection, faulty control input, or hardware issue.

Apart from this deviation, the overall operation of the IC 7447 and the 7-segment display is verified successfully.

Conclusion:

The experiment verifies the correct operation of IC 7447 in driving a common-anode 7-segment display for BCD inputs. Active-LOW outputs and ripple blanking were observed.